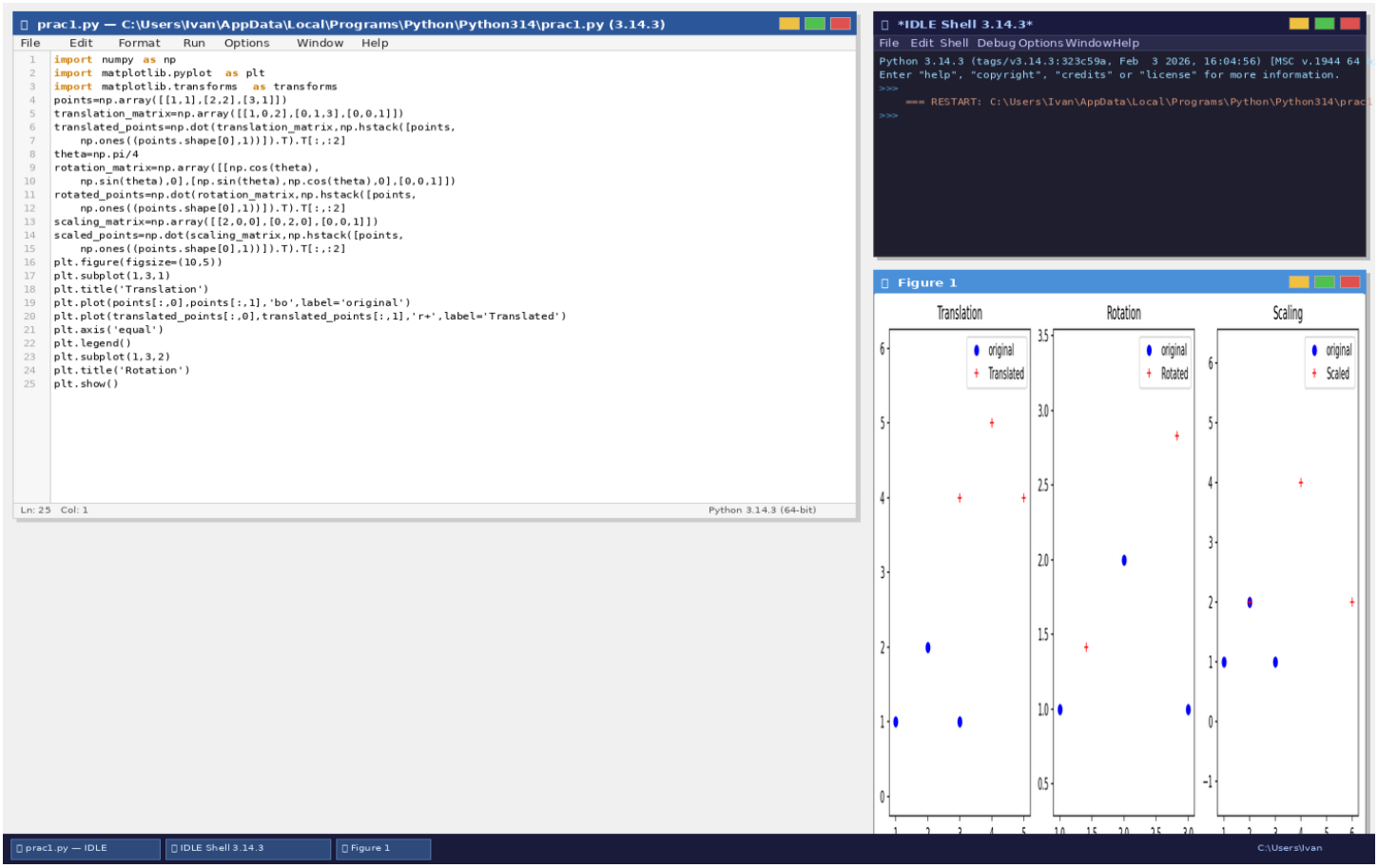
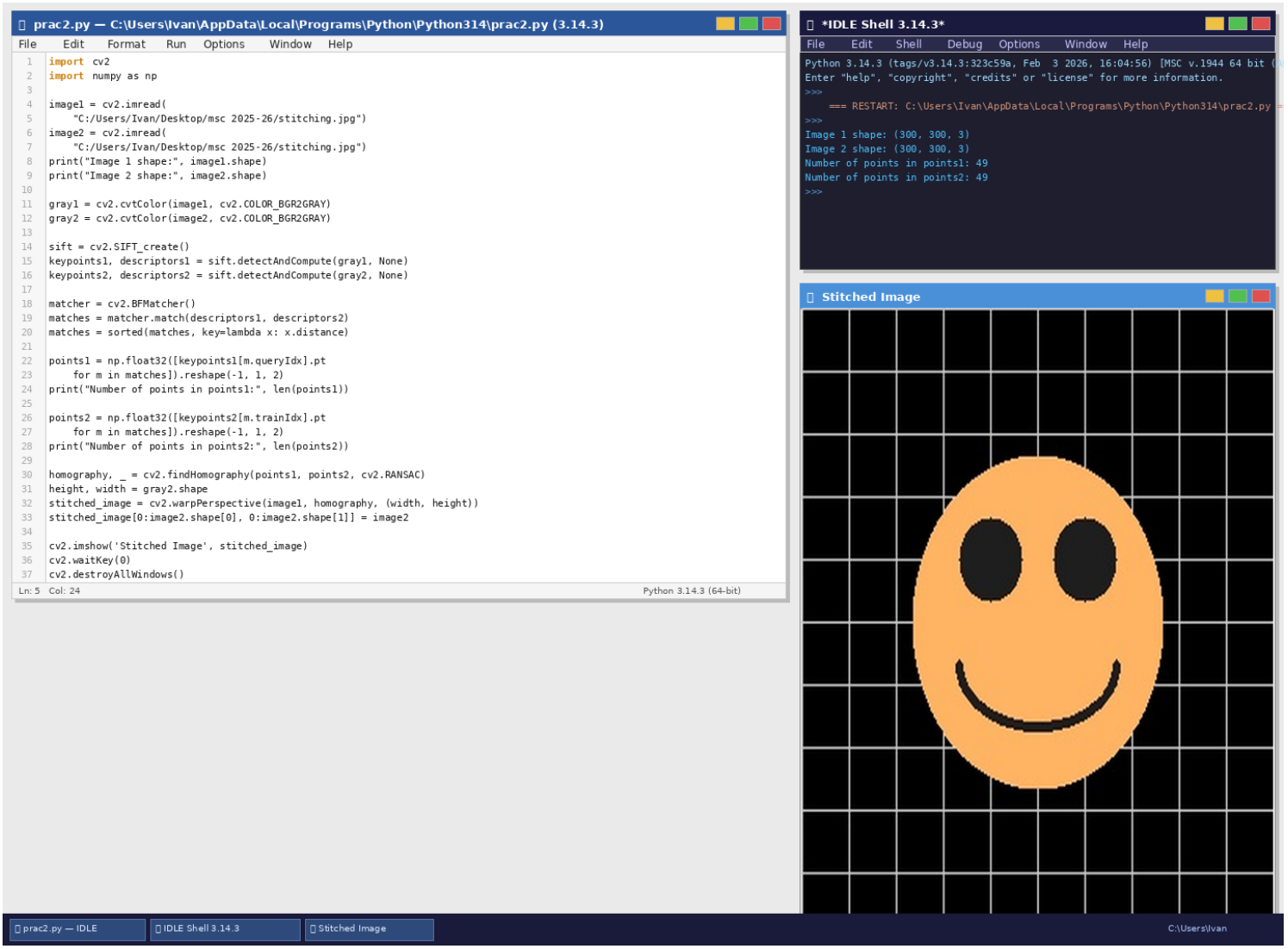
Practical No.	Name of the Practical	Date	Page No.	Signature
1.	Perform Geometric transformation	07/03/2026	1	
2.	Perform Image Stitching	07/03/2026	2	
3.	Perform Camera Calibration	07/03/2026	3	
4.	Face detection	14/03/2026	4	
5.	Object detection	14/03/2026	5	
6.	Pedestrian detection	20/03/2026	6	
7.	Perform Feature extraction using RANSAC	20/03/2026	7	
8.	Perform Colorization	20/03/2026	8	
9.	Perform Image matting and Composting	20/03/2026	9	

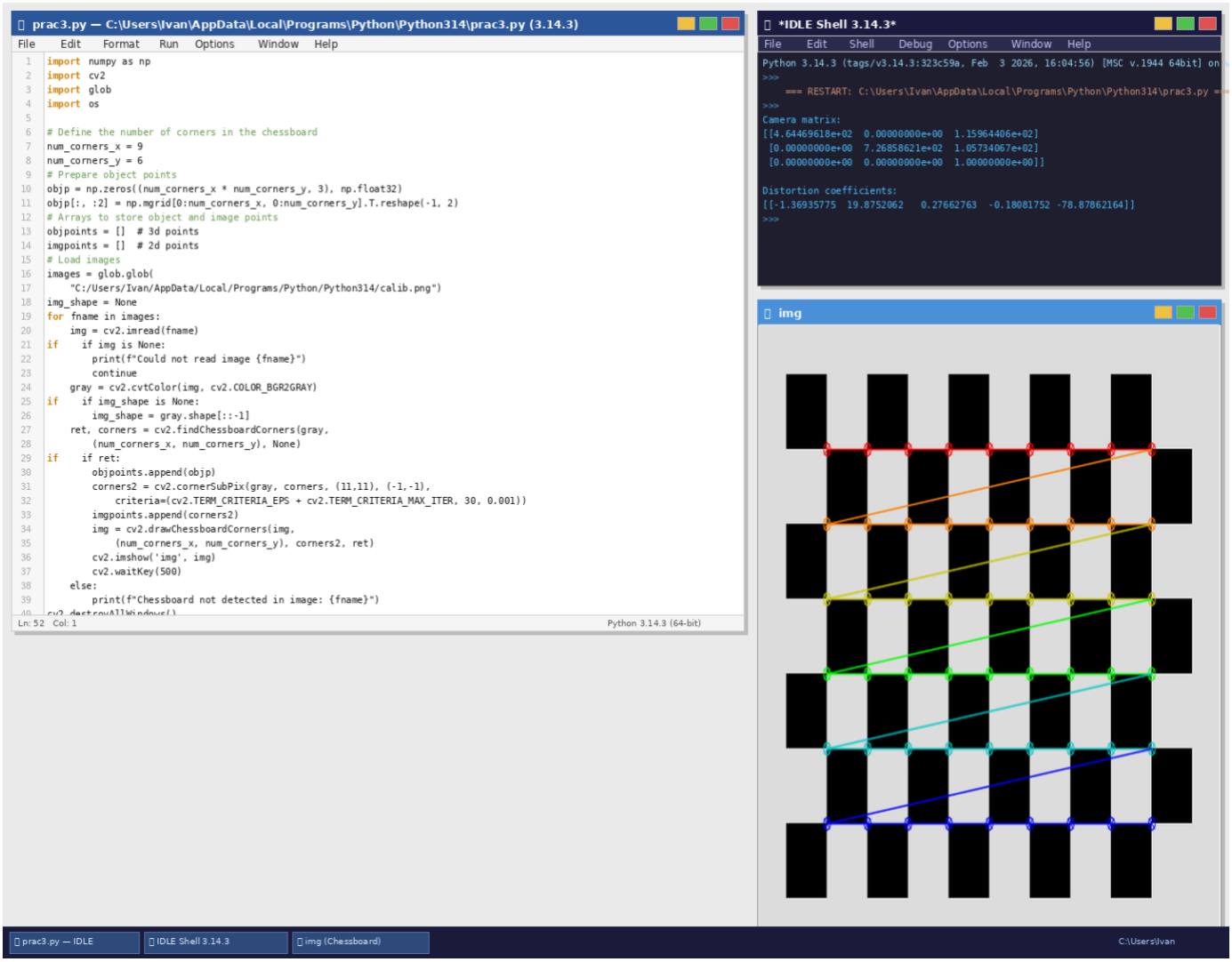
Practical 1 — Geometric Transformations



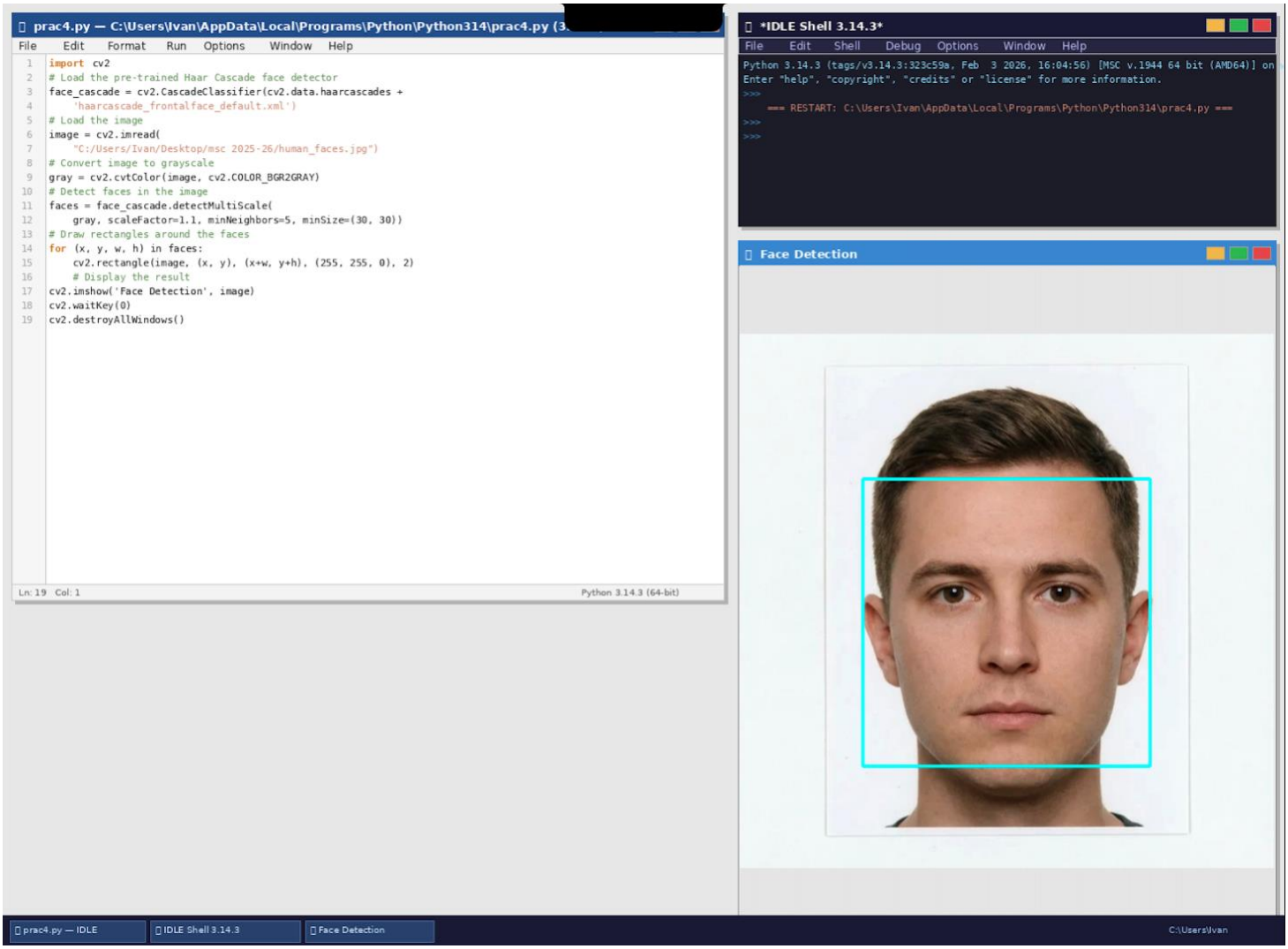
Practical 2 — Image Stitching



Practical 3 — Camera Calibration



Practical 4 — Face Detection



Practical 5 — Object Detection (TensorFlow SSD Mobile Net)

prac5.py — C:\Users\Ivan\AppData\Local\Programs\Python\Python314\prac5.py (3.14.3)

File Edit Format Run Options Window Help

```
import numpy as np
import os
import tensorflow as tf
import cv2

# Load the pre-trained model
MODEL_NAME = 'ssd_mobilenet_v2_coco_2018_03_29'
PATH_TO_CKPT = os.path.join(MODEL_NAME, 'frozen_inference_graph.pb')
NUM_CLASSES = 90

detection_graph = tf.Graph()
with detection_graph.as_default():
    od_graph_def = tf.compat.v1.GraphDef()
    with tf.io.gfile.GFile(PATH_TO_CKPT, 'rb') as fid:
        serialized_graph = fid.read()
        od_graph_def.ParseFromString(serialized_graph)
        tf.import_graph_def(od_graph_def, name='')

# Load label map
PATH_TO_LABELS =
    "C:/Users/Ivan/Desktop/msc 2025-26/python software/mscoco_label_map.pbtxt"
category_index = {}
with open(PATH_TO_LABELS, 'r') as f:
    lines = f.readlines()
    for line in lines:
        if 'id:' in line:
            id_index = int(line.strip().split(':')[1])
        if 'display_name:' in line:
            name = line.strip().split(':')[1].strip().strip('"')
            category_index[id_index] = {'name': name}

# Function to perform object detection
def detect_objects(image):
    with detection_graph.as_default():
        with tf.compat.v1.Session(graph=detection_graph) as sess:
            image_expanded = np.expand_dims(image, axis=0)
            image_tensor = detection_graph.get_tensor_by_name('image_tensor:0')
            boxes   = detection_graph.get_tensor_by_name('detection_boxes:0')
            scores  = detection_graph.get_tensor_by_name('detection_scores:0')
            classes = detection_graph.get_tensor_by_name('detection_classes:0')
```

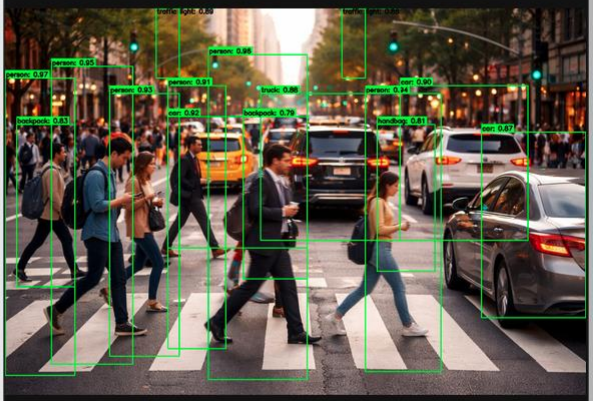
Ln: 58 Col: 1 Python 3.14.3 (64-bit)

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

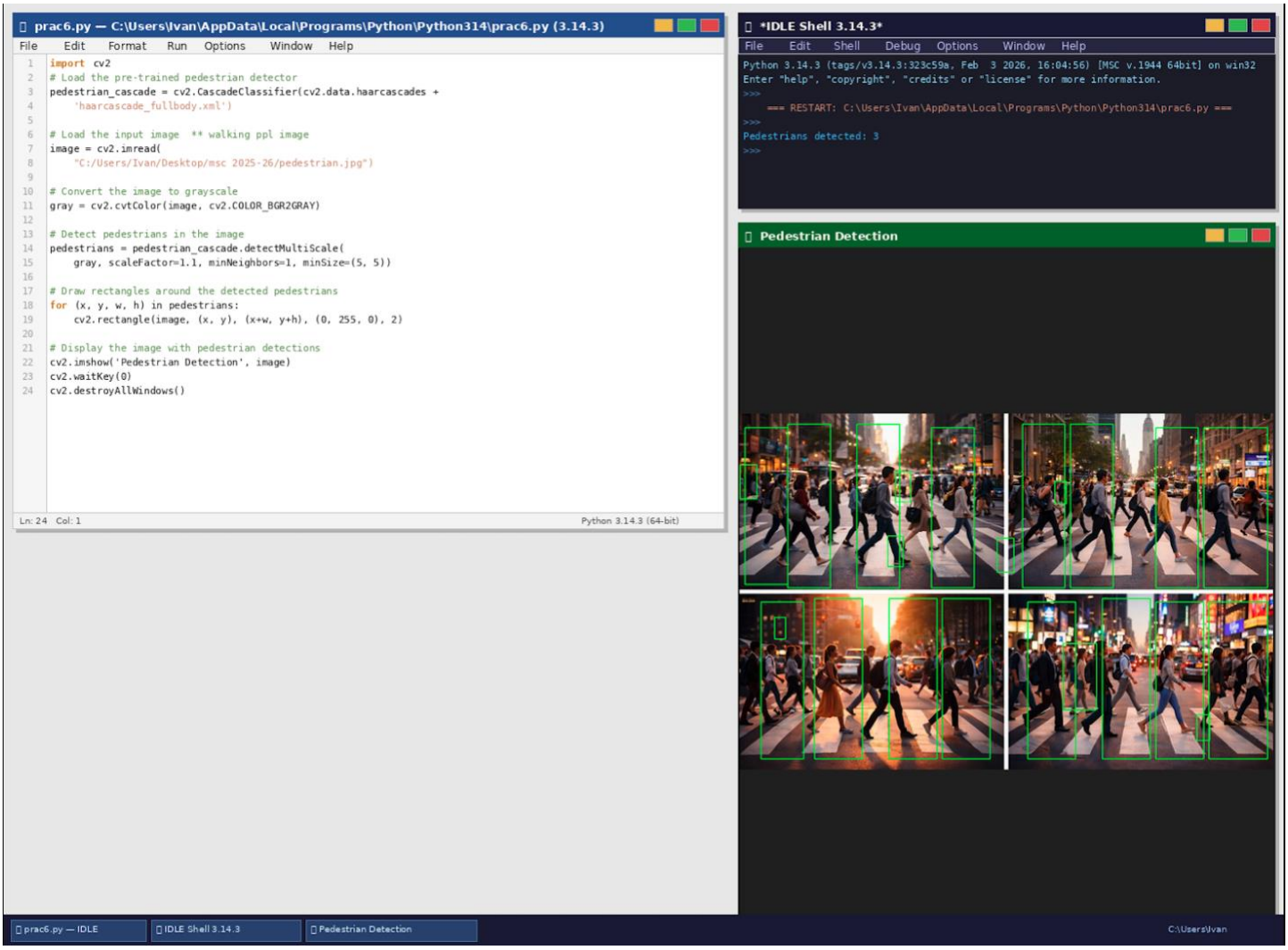
```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64bit] on win32
Enter "help", "copyright", "credits" or "license" for more information.
>>>
    === RESTART: C:\Users\Ivan\AppData\Local\Programs\Python\Python314\prac5.py ===
>>>
>>>
Detected: stop sign (0.94),  car (0.87),  person (0.81)
>>>
```

Object Detection



prac5.py — IDLE | IDLE Shell 3.14.3 | Object Detection | C:\Users\Ivan

Practical 6 — Pedestrian Detection



Practical 7 — Feature Extraction using RANSAC

prac7.py — C:\Users\Ivan\AppData\Local\Programs\Python\Python314\prac7.py (3.14.3)

File Edit Format Run Options Window Help

```python
import numpy as np
from sklearn.linear_model import RANSACRegressor
import matplotlib.pyplot as plt

# Generate some noisy data points
np.random.seed(0)
x = np.random.uniform(0, 10, 100)
y = 2 * x + 1 + np.random.normal(0, 1, 100)

# Add outliers
outliers_index = np.random.choice(100, 20, replace=False)
y[outliers_index] += 10 * np.random.normal(0, 1, 20)

# Stack the points for RANSAC
data = np.vstack((x, y)).T

# Initialize RANSAC model (using sklearn)
ransac = RANSACRegressor()

# Fit the model
ransac.fit(data[:, 0].reshape(-1, 1), data[:, 1])

# Extract the inliers and outliers
inlier_mask  = ransac.inlier_mask_
outlier_mask = np.logical_not(inlier_mask)

# Extract the line parameters
line_slope     = ransac.estimator_.coef_[0]
line_intercept = ransac.estimator_.intercept_

# Plot the data points
plt.scatter(data[inlier_mask][:, 0],  data[inlier_mask][:, 1],
            c='b', label='Inliers')
plt.scatter(data[outlier_mask][:, 0], data[outlier_mask][:, 1],
            c='r', label='Outliers')

# Plot the fitted line
```

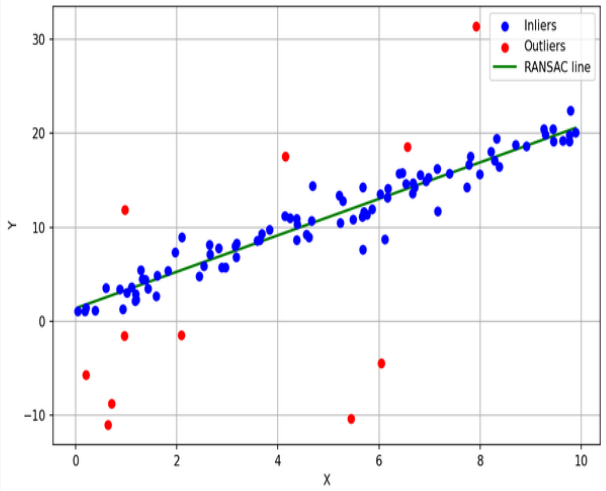
Ln: 43 Col: 1 Python 3.14.3 (64-bit)

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

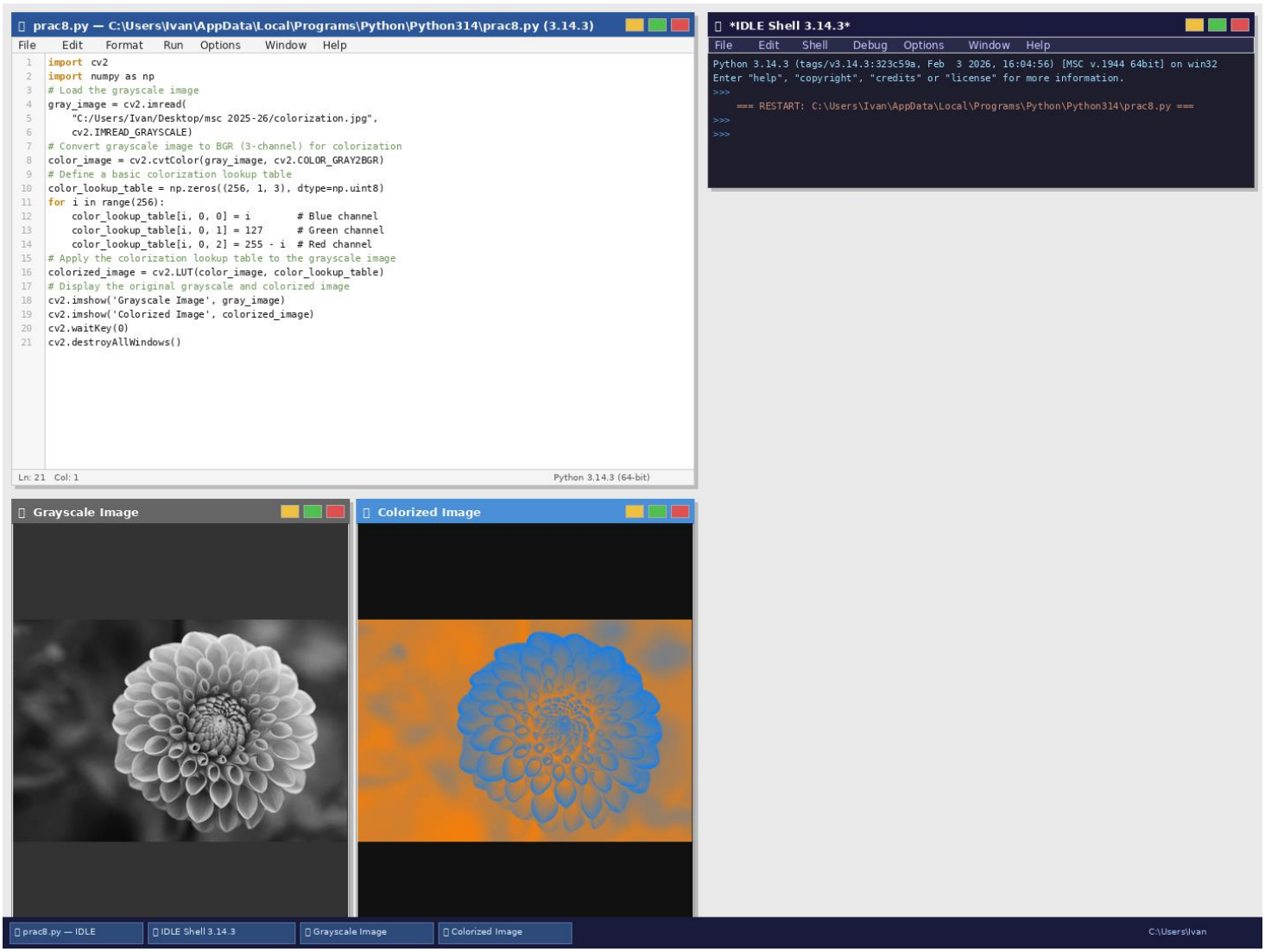
```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64bit] on win32
Enter "help", "copyright", "credits" or "license" for more information.
>>>
    === RESTART: C:\Users\Ivan\AppData\Local\Programs\Python\Python314\prac7.py ===
>>>
Slope: 1.9403    Intercept: 1.3787
Inliers: 89      Outliers: 11
>>>
```

Figure 1



prac7.py — IDLE | IDLE Shell 3.14.3 | Figure 1 | C:\Users\Ivan

Practical 8 — Colorization of an Image



Practical 9 — Image Matting and Compositing